Session Objectives

1. Explain the pathogenesis, pathophysiology, pathology and clinicopathology of **Benign Nephrosclerosis**.
2. Compare the pathogenesis, pathology, pathophysiology and clinicopathology of **Malignant Nephrosclerosis** with Benign Nephrosclerosis.
3. Contrast the various etiologies, pathogeneses, pathology, pathophysiology and clinicopathology of **Thrombotic Microangiopathies** (TTP and HUS)
4. Illustrate the pathogenesis, pathology, pathophysiology and clinicopathology of **Renal Artery Stenosis/Occlusion** and **Renal Ischemia/ Infarction.**

Benign Nephrosclerosis

- Renal pathology associated w/ **sclerosis** of renal **arterioles** & small arteries → **focal ischemia**
  - “Benign" because renal function is minimally affected, or proceeds to chronic kidney injury **slowly**
  - Distinguishes it from malignant nephrosclerosis (renal lesions of malignant hypertension) - **rapidly** progressive severe kidney damage

- Associated w/ **hypertension**s and also,
  - Advanced **age**, **African-Americans** > Caucasians
  - Incidence & severity ↑ by **Diabetes Mellitus**
- May also be seen in the absence of hypertension

Benign nephrosclerosis pathogenesis

- With **chronic** hypertension pathogenesis involves 2 processes
  - **1. Medial** & **intimal thickening** (hyalinization & thickened vessel walls from myofibroblastic tissue)
  - **2. Hyaline deposition** in small arterioles (**hyaline arterioloscerosis**) – caused by extravasation of plasma proteins through injured endothelium ↑deposition of basement membrane matrix

- Thickened vessel walls cause narrowed lumens: **Ischemia** leads to glomerulosclerosis & chronic tubulointerstitial injury

Malignant hypertension

- ~1-5% of hypertension patients show a **rapidly rising blood pressure**
  - Cause: Endothelial **injury**

- Populations at ↑ risk: **Males**, African-Americans

- Usually **superimposed on** individuals with **underlying** disorders, such as:
  - Benign hypertension - aka, essential hypertension, bp 130-139/80-89, usually treated with lifestyle modifications
  - Underlying chronic renal diseases like glomerulonephritis, reflux nephropathy…

Malignant hypertension evolution

- Pathogenesis: ↑ **permeability** of small vessels to fibrinogen & other plasma proteins, endothelial **injury, focal cell death** of cells of the vascular wall & **platelet** deposition
- A clinical scenario: **Diastolic** pressure **> 120 mm Hg**, papilledema, encephalopathy, HA, nausea, vomiting, visual problems, renal failure, CV problems→ **Medical emergency**

Thrombotic microangiopathies

- Spectrum of clinical syndromes due to **endothelial injury**, excessive **platelet** activation & aggregation, that notably includes the following:
  - Malignant hypertension
  - HUS – **hemolytic uremic syndrome**
    - Typical
    - Atypical
  - TTP – **thrombotic thrombocytopenic purpura**

- Result in the following: **Thrombi** in capillaries/arterioles create flow abnormalities – RBCs are sheared (microangiopathic hemolytic anemia, **MAHA**); thrombi cause microvascular **occlusions** which lead to tissue **ischemia**

Thrombotic microangiopathy - Typical HUS

- Other descriptors are epidemic, classic, diarrhea positive
  - **E.coli strain O157:H7** – intestinal infection usually from contaminated food such as ground **meat** produces a
    - **Shiga-like toxin -** trigger for endothelial **injury** & causes **platelet** activation & **thrombosis** within microvasculature
    - Most common cause in US
  - **Shigella dysenteriae**, serotype 1, is a specific type of Shigella bacteria that can produce a Shiga toxin & HUS (via contaminated food, water, infected persons)

  - Usual populations affected: **Children** & **elderly**
  - After **diarrhea,** sudden onset of **hematemesis, melena,** severe **oliguria, hematuria, MAHA, thrombocytopenia**, and sometimes **neurological** changes

Atypical HUS

- AKA Non-epidemic, diarrhea-negative
  - Patients **don't do as well** as typical HUS since underlying problem chronic and difficult to treat
  - Associated w/ **mutations** or **autoantibodies** of complement regulatory proteins
    - 1/2 of cases - inherited **Factor H deficiency** (uncontrolled complement activation)
    - Other patients have mutations in other complementary regulatory proteins (e.g., complement **Factor I** and **CD46**)

Thrombotic microangiopathy - TTP

- **Pentad** of fever, neurological symptoms, MAHA, thrombocytopenia, and renal failure (classic definition)
  - More **likely** to have **CNS** presentation**, renal** involvement in **50%**
  - Most common cause: Inactivating **autoantibodies** against **ADAMTS13** – regulates von Willebrand factor
    - More common in **women**, especially **< 40** years of age
    - Removal of autoantibodies with **plasma exchange**, FFP & **recombinant** ADAMTS13 provides functional ADAMTS13 in >80% of patients
  - ADAMTS13 **mutations** – less common cause; i.e. inherited is less likely than acquired

Renal artery stenosis

- **Unilateral** renovascular hypertension is responsible for **2-5% of hypertension**
  - Important to recognize since it is one of the (surgically) **curable** causes of hypertension
  - 90-95% of hypertension cases are **idiopathic**

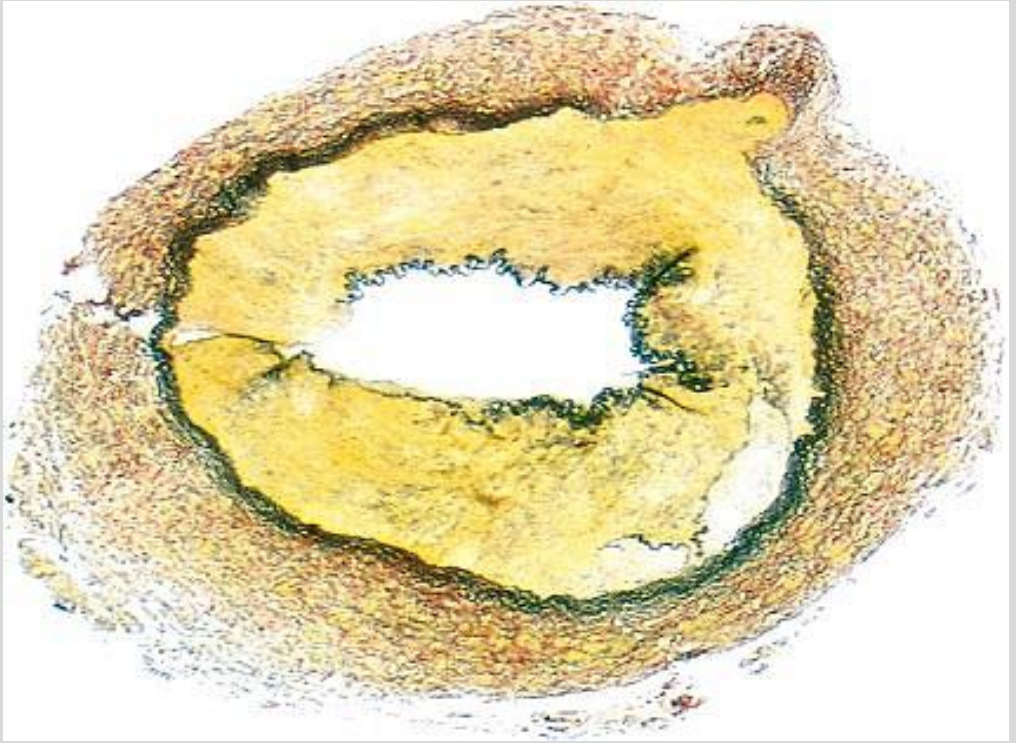
- Most common cause – **atheromatous plaque**, especially **males** w/ advancing **age** & **diabetes**
- 2nd most common cause – **fibromuscular dysplasia** of renal artery

- **Clinically** patient's resemble essential hypertension, occasional **bruit** over affected kidney, elevated plasma or renal vein **renin**, response to **ACE inhibitors**, **arteriography** localizes stenotic lesion
  - Arterioles of ischemic kidney usually protected from pressure
  - Contralateral nonischemic kidney may show more severe arteriolosclerosis

- Diffuse ischemic atrophy of kidney histopathologically shows: Shrunken kidney w/ sclerotic, crowded glomeruli, atrophic tubules, interstitial fibrosis & focal inflammation

Fibromuscular dysplasia histopathology

- Fibromuscular thickening may involve intima, media, or adventitia
- More common in **young women** (20 & 30 y.o.)



- Verhoeff–Van Gieson stained renal artery tissue
  - Elastin black, collagen red, other yellow
- Fibromuscular dysplasia of the renal artery, **medial hyperplasia type** (yellow)
- Tunica media shows marked fibrous thickening & stenotic lumen

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Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2026, Ch. 20

Renal infarction

- Kidneys have **limited collateral** circulation from extra-renal sites
  - Most infarcts are **pale** or **anemic**
- Usually due to **emboli**
  - Major source of emboli is mural thrombosis of **left atrium** & **ventricle** from MI
  - Sometimes caused by advanced **atherosclerosis**, or polyarteritis nodosa
- Usually clinically **silent**, sometimes may have CVA tenderness
- Coagulative necrosis occurs: Renal infarcts are **wedge shaped** w/ **base** against **cortical** surface & apex pointing toward medulla – may undergo fibrosis and form **V-shaped** scar

Other renal vascular diseases

- **Atherosclerotic** ischemic renal disease – **bilateral** renal artery disease causes **chronic** ischemia
  - Typically, older patients w/ renal **insufficiency**, sometimes **w/o hypertension**
  - Dx by arteriography
  - Revascularization prevents renal function decline

- **Athero-embolic** renal disease – athero-emboli from aorta or renal artery travel into intrarenal vessels
  - Usually in **older** adults w/ severe atherosclerosis, especially after **surgery** on abdominal aorta, aortography, etc.

- **Sickle cell** nephropathy – **hematuria** & ↓ **concentrating** ability are major manifestations because of sickling in hypertonic hypoxic environment of **renal medulla**
  - Patchy **papillary necrosis** may develop